Disclaimer

Environment: This notebook has been tested with:

- Julia v1.12.x
- JuMP.jl v1.30.x
- HiGHS v1.24.1

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Case 3: Application

Preamble: classification of resolution methods according to the consideration of the decision-maker's preferences:

- a posteriori methods
- a priori methods
- interactive methods

Aim: computing interactively $y \in Y_N$ for

$$\begin{array}{rl} \text{maximize} & f(x) = \left[f_1(x), \ldots, f_d(x)\right] \\ \text{subject to} & x \in X \end{array}$$

with **the augmented weighted Tchebycheff scalarizing function**.

$\longrightarrow$ explanation of this function with the Pluto notebook...

Augmented Weighted Tchebycheff

```
Compute one nondominated point by application of the
Tchebycheff theory.

In  Steuer, R.E., Choo, EU. An interactive weighted
Tchebycheff procedure
    for multiple objective programming. Mathematical
Programming 26, 326–344 (1983).
```

The scalarizing function $s(X, \lambda, r^*, \rho)$:

$$\begin{array}{ll} \text{minimize} & \left[\max\limits_{k=1,...,d} \left\{ \lambda_k \big(r_k^* - z_k(x) \big\} + \rho \sum\limits_{k=1}^{d} \big(r_k^* - z_k(x) \big) \right] \\ \text{subject to} & x \in X \end{array}$$

where

- $z_k(x)$: value of objective k for solution x
- r_k^* : value of the reference point for objective k
- λ_k : weight associated with objective k in the scalarization
- ρ : small augmentation coefficient

The `max` inside makes this formulation not directly usable by a standard MILP solver.

However this expression can be linearized as follows: an auxiliary variable α, which upper-bounds all expressions inside the max, is introduced, and α plus the augmented term are minimized.

Linearizing the scalarization function $s(X, \lambda, r^*, \rho)$ when the problem under consideration is mo01UKP leads to the following formulation:

$$\begin{array}{lll} \min & \alpha + \rho \sum\limits_{k=1}^{d} \big(rp_k - z_k(x) \big) & \\ \text{s.t.} & \alpha \geq \lambda_k \big(rp_k - z_k(x) \big) & \forall k = 1, \ldots, d \\ & z_k(x) = \sum\limits_{j=1}^{n} p_{kj} \, x_j & \forall k = 1, \ldots, d \\ & \sum\limits_{j=1}^{n} w_j \, x_j \leq c & \\ & \alpha \geq 0 & \\ & x_j \in \{0, 1\} & \forall j = 1, \ldots, n \end{array}$$

The term α captures the maximum weighted deviation between the reference point rp and the current solution $z(x)$ across all objectives, while the additional term prevents generating weakly dominated points by slightly penalizing the sum of deviations, thereby ensuring the minimizer is a properly (strictly) nondominated point.

Implement the augmented Tchebycheff scalarizing function on the mo01UKP:

Build the JuMP model implementing the augmented weighted Tchebycheff scalarization (linearized formulation given here) for the mo01UKP with d objectives, using a reference point rp set to an utopian point y^U (already discussed before) and a given weight vector λ (e.g. $\lambda = (\frac{1}{d}, \ldots, \frac{1}{d})$. Solve with HiGHS and display the nondominated point obtained.

The multi-objective 01 Unidimensional Knapsack Problem as context

A generator of mo01UKP instance

In [ ]:

```
using Random

function generate_mo01UKP(n = 10, d = 2, max_ci = 100, max_wi = 30)
    Random.seed!(123)

    p = rand(1:max_ci,d,n)      # c_i \in [1,max_ci]   # profits
    w = rand(1:max_wi,n)        # w_i \in [1,max_wi]   # weight
    c = round(Int64, sum(w)/2)                         # capacity

    return p, w, c
end
```

Our instance test

In [ ]:

```
function test_mo01UKP()

    p1 = [ 13, 10,  3, 16, 12, 11,  1,  9, 19, 13 ] # profit 1
    p2 = [  1, 10,  3, 13, 12, 19, 16, 13, 11,  9 ] # profit 2
    p3 = [ 17, 19,  3,  6, 12, 17, 18,  6,  6, 14 ] # profit 3

    w  = [  4,  4,  3,  5,  5,  3,  2,  3,  5,  4 ] # weight
    c  = 19                                         # capacity

    return [p1'; p2'; p3'] , w , c
end
```

Packages to use

In [ ]:

Compute a reference point

In [ ]:

Optimize a augmented weighted Tchebycheff problem

In [ ]:

Example of computing one point $y \in Y_N$ with $s(X, \lambda, r^*, \rho)$

In [ ]: